VeriTriage — FINREG Methodology

Track 5 AML · 397,853 ČNB transactions / 5,493 clients · load 75.9 ms · deep analysis 4.9 ms · 0 LLM tokens for arithmetic

Table 1 — Deterministic 0–100 Priority Formula

Component	Points	Rule	ALERT-045
Base risk	10/25/40	low / medium / high client risk	40
Velocity	0–30	`clamp(round((Actual/Expected − 1) × 15), 0, 30)`	11 (1.71x)
PEP	0/25	PEP flag, § 9a(2)(c) Act 253/2008	25
Structuring / typology	0–20	72h sub-threshold + HHI + round amount	20 (13 tx)
Channel	0/10	remote onboarding + velocity, Decree 67/2018 § 8(2)	0
Total · FP < 40 / REVIEW ≥ 40			**96**

ALERT-045: B40 + V11 + P25 + S20 + C0 = 96/100 · PEP REVIEW.

Table 2 — Five Quantitative Typology Vectors

Vector	Definition	ALERT-045
Velocity / count	Actual vs expected 2Q2026 volume and count; `System.Decimal`, MidpointRounding.ToEven	1.71x / 1.71x · 68,262.30 / 39,819.67 CZK · 12 / 7
72h structuring	Maximum rolling 72h cluster of sub-threshold transfers < 25,000 CZK	13 transfers / 95,511.91 CZK · 2026-07-11..07-14
Counterparty HHI	Σ(volume share²) × 10,000; shared client links	HHI 916 · primary ACZ00005 · 3 linked clients
Round amount	Share of amounts divisible by 1,000 or 500	0.00%
In/out pass-through	incoming / outgoing in client window	0.6495

Table 3 — Statutory Escalation & 4-Eyes Matrix

Originator	Required approver	Statutory basis	Server behavior
L1 Analyst	L2 Senior AML Specialist	§ 6(1)(a) Act 253/2008; Decree 67/2018 § 7	4-eyes check
L2 Senior Specialist	MLRO / designated contact	§ 9a(2)(c), § 22(1)	PEP/high-risk → MLRO
MLRO	Statutory board member	§ 9a(2)(c), § 18	senior management approval
Any originator	Same person	Decree 67/2018 § 12	**HTTP 400 4-EYES VIOLATION**
Any	Independent approver	§ 12 separation of duties	approved + audit event

The invariant is server-side: /approve compares executive_person and approver; the UI self-approval trap cannot bypass it.

Page 2 — Table 4: Dual-Mode Architecture

Mode	Engine	Tokens	Latency	Output
Deep Analysis	100% C#: vectors, 5 gates, priority	0	4.9 ms	Markdown brief + QUANTITATIVE_FORENSIC_BRIEF dossier
Statutory Investigation	Bedrock eu-central-1 + legal corpus + C# diff	1 call	~2.7 s (snapshot)	STATUTORY_FAU_OPO_LEGAL_MEMO dossier

Investigation receives deterministic vectors as immutable constraints; C# verifier overrides priority/volume and enforces 4-eyes. Legal dossiers persist in `data/legal_dossiers.json`, are hash-bound to the file audit ledger, and are signed with **NIST P-256 ECDSA** (SPKI public key in JSON). HTML export is print-ready; Markdown-only is approved.

Honest boundaries: 72h cluster is 13/95,511.91 CZK (not the earlier 5/23,778 subset). Current Bedrock call falls back to snapshot because the host .NET/AppleCrypto stack raises “bad certificate format”; the 37 ms figure was a prior run; measured load today is 75.9 ms. CSV 45 rows / 11 columns unchanged. Documents: FINAL_REPORT.md · DETERMINISTIC_LOGIC_REPORT.md · STANDARDS_ALIGNMENT.md.